

LOGICAL STATEMENTS

Chapter 4

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RELATIONAL EXPRESSIONS

- Relational operators are used to perform a True/False test:
 - Statement is TRUE – result is 1
 - Statement is FALSE – result is 0
 - Relational operators:
 - > greater than
 - < less than
 - >= greater than or equal to
 - <= less than or equal to
 - == equality
 - ~= inequality
- To check more than one condition use logical operators:
 - | or
 - & and
 - ~ not
- Remember code is executed from left to right
 - Important when thinking about logic

if STATEMENT

- Think about being on a budget and shopping for a new shirt
- You go to the mall and find two shirts that you like
- One shirt is \$30 and the other is \$100
- You have a choice to make:
 - Buy the cheaper shirtOR
 - Buy the expensive shirt
- This gives two potential paths for you to take
- **if** statements are designed for these types of issues
 - TRUE/FALSE decision(s)
 - Multiple paths

if STATEMENT

- How to set up an if statement:
if relational expression
calculations/action statements
end
- relational expressions used to do TRUE/FALSE test
 - If TRUE calculations/action statements are executed
 - Example actions: plotting, fprintf, load, save, equations, etc.
 - If FALSE skip to end statement

EXAMPLE

- A retail store is updating prices to reflect a sale
- All shirts over \$35 are 25% off
- Determine if a shirt should be discounted and if so, display sales price

```
price=input('Enter the price of the shirt: ');  
if price>35  
    sale=price*.75;  
    fprintf('Sale price at 25 percent off is:  
$%.2f\n',sale)  
end
```

Output:

```
>>Enter the price of the shirt: 40  
>>Sale price at 25 percent off is: $30.00
```

IF-ELSE STATEMENTS

- Does the situation require two separate paths with different actions?
 - **if else** statements take care of this
- Similar to the if statement except now we have two separate actions
- Form:

```
if condition
    Action #1
else
    Action #2
end
```
- condition is true -> action #1 is executed
- condition is false -> action #2 is executed
- Think of this as an either/or situation
 - Either do action 1
 - Or do action 2

EXAMPLE

- Going back to retail sales price example
- Now add the requirement that if the shirt is not on sale a message will be displayed

```
price=input('Enter the price of the shirt: ');  
if price>35  
    sale=price*.75;  
    fprintf('Sale price at 25 percent off is:  
$%.2f\n',sale)  
else  
    fprintf('Shirt is not on sale.\n')  
end
```

Output:

Enter the price of the shirt: 10

Shirt is not on sale.

Enter the price of the shirt: 50

Sale price at 25 percent off is: \$37.50

MORE THAN 2 PATHS

- Two ways to handle more than 2 paths:

- Nested if-else statements

- Can be hard to read
- Lots of end statements

```
if conditionA
    action A
else
    if conditionB
        action B
    else
        if conditionC
            action C
        else
            action final
        end
    end
end
```


MORE THAN 2 PATHS

- Simpler, cleaner way: **if-elseif**
 - Eliminates the multiple end statements
 - Easier to read and follow

```
if conditionA
```

```
    action A
```

```
elseif conditionB
```

```
    action B
```

```
elseif conditionC
```

```
    action C
```

```
else
```

```
    action final
```

```
end
```

Notice this is e l s e

switch STATEMENT

- Have multiple paths but don't need relational expressions:
- Use **switch**
- General form:

```

switch variable
  case case_exp1
    action 1
  case case_exp2
    action 2
  case case_exp3
    action 3
  % etc
  otherwise
    action n
end

```

```

switch - reserved word
variable - variable to be tested
case_exp1 - test variable against this
             if true, do action 1
case_exp2 - first condition failed, test
             this condition. True-do action 2
case_exp3 - second condition failed... etc

```

- **otherwise** action done for all other possible values/strings not previously listed

menu

- **menu** provides way to interactively collect info from user
- Pop-up with option buttons is displayed
- User selects a button and a **value** is returned
 - First button selected returns a 1
 - Second button selected returns a 2
 - Etc.
- `if-elseif` or `switch` then used to select between different paths
- Cannot use this with automated codes



menu

```
choice = menu('instructions','option1','option2','option3')
```

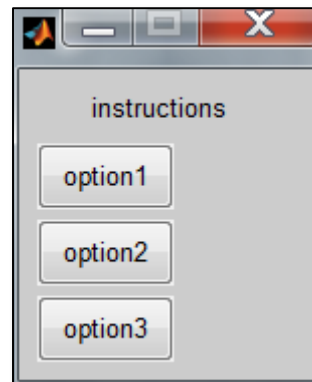
choice - the returned value is stored in this variable

instructions - text/instructions for the user

option1 - text that appears on top button. Returns a value of 1

option2 - text that appears on second button. Returns a value of 2

option3 - text that appears on the third button. Returns a value of 3



EXAMPLE

- What we want is a script that prompts the user for a value of a variable x .
- Use the **menu** function to present choices between 'sin(x)', 'cos(x)', and 'tan(x)'
- The script will print whichever function of x the user chooses.
- We will do this using two methods to make the choice:
 - **if-else**
 - **switch**

SOLUTION USING `if-else`

```
% Prints either sin, cos, or tan of x
% uses the menu function to choose

x = input('Enter a value for x: ');
choice = menu('Choose a function', 'sin','cos','tan');
if choice == 1
    fprintf('sin(%.1f) is %.1f\n', x, sin(x))
elseif choice == 2
    fprintf('cos(%.1f) is %.1f\n', x, cos(x))
elseif choice == 3
    fprintf('tan(%.1f) is %.1f\n', x, tan(x))
else
    disp('Error!')
end
```

SOLUTION USING `switch`

```
% Prints either sin, cos, or tan of x
% uses the menu function to choose
x = input('Enter a value for x: ');
choice = menu('Choose a function', 'sin','cos','tan');
switch choice
    case 1
        fprintf('sin(%.1f) is %.1f\n', x, sin(x))
    case 2
        fprintf('cos(%.1f) is %.1f\n', x, cos(x))
    case 3
        fprintf('tan(%.1f) is %.1f\n', x, tan(x))
    otherwise
        disp('Error!')
end
```